

# Daniel Rho

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## Research Interests

computer vision, neural rendering, 3D / 4D representation learning, and inverse physics

## Education

### University of North Carolina at Chapel Hill

Ph.D. in Computer Science

North Carolina, United States

Aug. 2024 - Present

- Advisor: [Roni Sengupta](#)

### Sungkyunkwan University

MSE in Artificial Intelligence

Seoul, Korea

Sep. 2020 - Aug. 2022

- Thesis: "Neural Residual Flow Fields for Efficient Video Representations" (Advisor: [Jong Hwan Ko](#), Co-advisor: [Eunbyung Park](#))

### Sungkyunkwan University

Bachelor of Economics & BSE in Computer Science and Engineering

Seoul, Korea

Mar. 2014 - Aug. 2020

- Major GPA (Computer Science and Engineering): 4.44 / 4.5 (top 3%)
- Dean's List, 2018

## Publications

### CONFERENCE PUBLICATIONS

#### MonoPhysics: Estimating Geometry, Appearance, and Physical Parameters from Monocular Videos

**Daniel Rho**, Jun Myeong Choi, Matthew Thornton, Biswadip Dey, Roni Sengupta

**NeurIPS 2026** - *Advances in Neural Information Processing Systems*

#### ProJo4D: Progressive Joint Optimization for Sparse-View Inverse Physics Estimation

**Daniel Rho**, Jun Myeong Choi, Biswadip Dey, Roni Sengupta

**TMLR 2026** - *Transactions on Machine Learning Research*

#### NFL-BA: Improving Endoscopic SLAM with Near-Field Light Bundle Adjustment

Andrea Dunn Beltran\*, **Daniel Rho**\*, Marc Niethammer, Roni Sengupta

**NeurIPS 2025** - *Advances in Neural Information Processing Systems*

#### F-3DGS: Factorized Coordinates and Representations for 3D Gaussian Splatting

Xiangyu Sun, Joo Chan Lee, **Daniel Rho**, Jong Hwan Ko, Usman Ali, Eunbyung Park

**ACM MM 2024** - *Proceedings of the 32nd ACM International Conference on Multimedia*

#### Compact 3D Gaussian Representation for Radiance Field

Joo Chan Lee, **Daniel Rho**, Xiangyu Sun, Jong Hwan Ko, Eunbyung Park

**CVPR 2024 (highlight)** - *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*

#### Coordinate-Aware Modulation for Neural Fields

Joo Chan Lee, **Daniel Rho**, Seungtae Nam, Jong Hwan Ko, Eunbyung Park

**ICLR 2024 (spotlight)** - *International Conference on Learning Representations*

#### Mip-Grid: Anti-aliased Grid Representations for Neural Radiance Fields

Seungtae Nam, **Daniel Rho**, Jong Hwan Ko, Eunbyung Park

**NeurIPS 2023** - *Advances in Neural Information Processing Systems*

#### FFNeRV: Flow-Guided Frame-Wise Neural Representations for Videos

Joo Chan Lee, **Daniel Rho**, Jong Hwan Ko, Eunbyung Park

**ACM MM 2023** - *Proceedings of the 31th ACM International Conference on Multimedia*

#### Masked Wavelet Representation for Compact Neural Radiance Fields

**Daniel Rho**\*, Byeonghyeon Lee\*, Seungtae Nam, Joo Chan Lee, Jong Hwan Ko, Eunbyung Park

**CVPR 2023** - *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*

#### Regression to Classification: Waveform Encoding for Neural Field-Based Audio Signal Representation

TaeSoo Kim\*, **Daniel Rho**\*, Gahui Lee, JaeHan Park, Jong Hwan Ko

**ICASSP 2023** - *IEEE International Conference on Acoustics, Speech and Signal Processing*

#### Neural Residual Flow Fields for Efficient Video Representations

**Daniel Rho**, Junwoo Cho, Jong Hwan Ko, Eunbyung Park

**ACCV 2022** - *Proceedings of the Asian Conference on Computer Vision*

Streamable Neural Fields

Junwoo Cho\*, Seungtae Nam\*, **Daniel Rho**, Jong Hwan Ko, Eunbyung Park  
ECCV 2022 - Proceedings of the European Conference on Computer Vision

NAS-VAD: Neural Architecture Search for Voice Activity Detection

**Daniel Rho**, Jinhyeok Park, Jong Hwan Ko  
Interspeech 2022 - Proceedings of Interspeech

PREPRINTS

Stop Holding Your Breath: CT-Informed Gaussian Splatting for Dynamic Bronchoscopy

Andrea Dunn Beltran, **Daniel Rho**, Aarav Mehta, Xinqi Xiong, Raúl San José Estépar, Ron Alterovitz, Marc Niethammer, Roni Sengupta  
arXiv preprint arXiv:2604.28179 (2026)

Understanding Contrastive Learning Through the Lens of Margins

**Daniel Rho**, TaeSoo Kim, Sooill Park, Jaehyun Park, JaeHan Park  
arXiv preprint arXiv:2306.11526 (2023)

Professional Experience

Graduate Research Assistant

University of North Carolina at Chapel Hill

Chapel Hill, NC, USA

Aug. 2024 - Present

Research Intern

Lenovo Research

NC, USA

May. 2025 - July. 2025

Research Engineer

AI Tech Lab, KT

Seoul, Korea

Jul. 2022 - Jun. 2024

Graduate Research Assistant

IRIS LAB, Sungkyunkwan University

Seoul, Korea

Sep. 2020 - Jul. 2022

Undergraduate Research Assistant

IRIS LAB, Sungkyunkwan University

Seoul, Korea

Jun. 2019 - Aug. 2020

Research Projects

Next-gen 3D Modeling of Endoscopy Videos

National Institutes of Health, USA

USA

2024-2026

Deep Learning Techniques for Multi-Intelligence using Drones

Ministry of Science and ICT, Korea - TRKO202300005241

Korea

Jan. 2021 - Dec. 2021

Deep Neural Network Based Real-Time Accurate Voice Source Localization using Drones

Ministry of Science and ICT, Korea

Korea

Jun. 2019 - Dec. 2020

Awards, Honors and Scholarships

Jan. 2021 **First Place & Ministerial Award**, Artificial Intelligence Grand Challenge, Ministry of Science and ICT

Korea

Fall 2020 **Sungkyun Honorable Scholarship (Fall 2020 - Spring 2022)**, Sungkyunkwan University

Korea

Jun. 2019 **Third Place**, Artificial Intelligence Grand Challenge, Ministry of Science and ICT

Korea

Fall 2019 **Academic Excellence Scholarship**, Sungkyunkwan University

Korea

Fall 2018 **Academic Excellence Scholarship**, Sungkyunkwan University

Korea

Academic Services

Conference Reviewer

CVPR 2024-2026, ICCV 2025, ECCV 2026, WACV 2026, NeurIPS 2024-2025, ICML 2025, ICLR 2025-2026, ACM MM 2024-2025, ACCV 2027

Journal Reviewer

TPAMI 2026

## Patents

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### “A Method for Inferring of Generating Direction of Sound Using Deep Network and an Apparatus for the Same”

Application No.: 10-2020-0032737

*Korea*

*2020*

## Skills

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### Programming

CUDA, Python (PyTorch, TensorFlow), C/C++

## Extracurricular Activities

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### Teaching Assistant

Sungkyunkwan University (SKKU)

- Operating Systems (Fall 2020)
- Basic data structures and algorithms (Spring-Fall 2019)

*Korea*